

Sprint 1 Documentation – Project Setup, REST API, DAL

Team: Sebastian and Felix

Repository: <https://github.com/BunnySupreme/SWEN3Project>

Sprint duration: 08.10.2025 – 16.10. 2025

Estimated workload: 10 h / person / week

1. Sprint Goal (from assignment)

- Setup .NET/C# backend project ("paperless" service)
- Provide REST API endpoints (code-first, defined by team)
- Integrate ORM (EF Core) with PostgreSQL using repository pattern
- Provide unit tests for DAL (mock DB)
- Initial docker-compose.yml with REST server + database

2. Implemented Functionality

2.1 Backend Project Setup

REST server paperless reachable via <http://localhost:5001>

2.2 Entities and Database Mapping

Entity: Document

Fields: Id, Title, Summary, Tags, UploadedAt

EF Core used with repository pattern:

- IDocumentRepository interface in Paperless.DAL.Repositories
- Implementation DocumentRepository in Paperless.DAL.Repositories
- DbContext: DataContext configured for PostgreSQL

2.3 REST Endpoints

- GET /api/documents: list documents
- POST /api/documents: post document (currently only metadata in a json)
- GET /api/documents/{id}: get document by ID
- POST /api/documents/{id}: update document metadata
- DELETE /api/documents/{id}: delete document

3. Architecture (Sprint 1)

- API layer: DocumentController
- Business layer: DocumentService
- Data access layer: DocumentRepository
- Database: PostgreSQL container from `docker-compose.yml`

Refer to Architecture.md for diagrams.

4. Testing

Unit tests implemented in project Paperless.UnitTests:

- DocumentServiceTests

Database mocked using Moq

All tests passing at the end of Sprint 1

5. Problems

Docker networking between API and PostgreSQL. After debugging it turns out Docker could not get the password to access and update the DB and it could not apply the migrations. Turns out the whole image has to be rebuilt, not just the container, after file change.

6. Workload (Estimate)

Team Member	Week / Sprint	Estimated Hours	Main Tasks
Sebastian	Sprint 1	10	CI/CD pipeline, Dockerfile, docker compose yaml, DAL, Configuration, ReadMe
Felix	Sprint 1	10	Project Setup, Github Setup, Controller, Endpoints, BL, DTOs